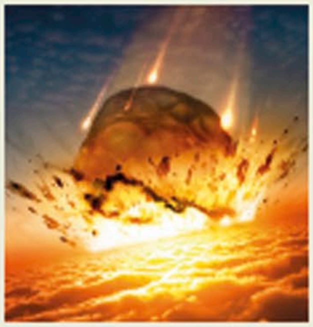
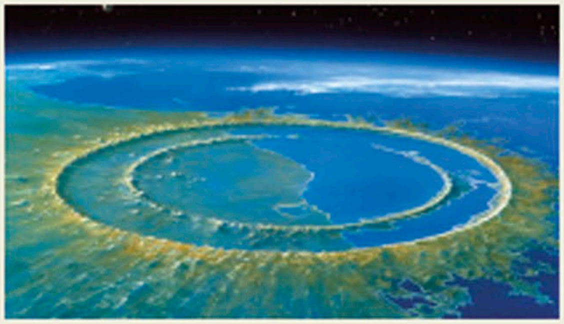
1980–1991 ALVAREZ HYPOTHESIS



Artist's depiction of the asteroid crashing into Earth

American scientists Luis Alvarez and his son Walter Alvarez found high levels of iridium—an element common in asteroids but not on Earth—in rocks that were 65 million years old. This led them to propose that dinosaurs died out at that time because of an asteroid impact. This would have produced enough dust to block out the Sun and cause major climate change.



Chicxulub crater
A 65-million-year-old crater at Chicxulub, in Mexico's Yucatán Peninsula, provided further evidence for the Alvarez hypothesis. Discovered in 1991, the crater has a diameter of 111.8 miles (180 km).

Artist's impression of the Chicxulub crater

1984

First untethered space walk

NASA astronauts Bruce McCandless and Robert Stewart left the space shuttle *Challenger* for a space walk—the first untethered space walk for any astronaut—using a Manned Maneuvering Unit (MMU). The MMU was a jet pack containing 26 lb (11.8 kg) of nitrogen fuel, which powered 24 small jet thrusters to move each astronaut through space without a cable tether keeping them attached to the spacecraft.

Astronaut Bruce McCandless on his spacewalk

1985

1983

Handheld cellular phone

More than a decade after a prototype was unveiled (see p.215), the first portable cellular mobile phone—the Motorola DynaTAC 8000X—went on sale. It was priced at $3995 and weighed 1.74 lb (790 g). It had a 10-hour charge on its battery, and gave its users a talk-time of up to 30 minutes. DynaTAC phones would stay on sale until 1994.

Motorola DynaTAC 8000X

1984

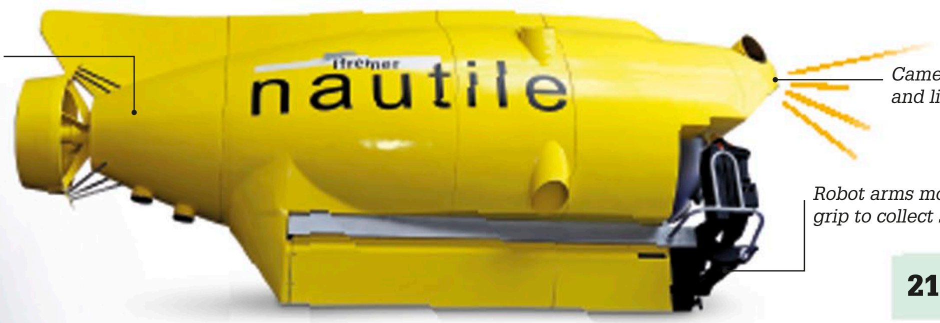
HIV identified

American biomedical researcher Robert Gallo and French virologist Luc Montagnier announced the discovery of HIV (human immunodeficiency virus), which is responsible for the deadly disease AIDS (Acquired Immune Deficiency Syndrome). HIV attacks the body's immune system and weakens a person's ability to fight disease and infections.

1984

Submersible *Nautile* launched

The French deep-sea submersible *Nautile* was launched in 1984. Capable of diving up to 19,685 ft (6,000 m) below sea level, it later filmed the wreckage of the British ship *Titanic* 12,467 ft (3,800 m) underwater, and helped recover more than 1,800 items from the wreck. *Nautile* also salvaged flight recorders of sunken aircraft.



Titanium hull protects passengers.

Cameras and lights

Robot arms move and grip to collect samples.